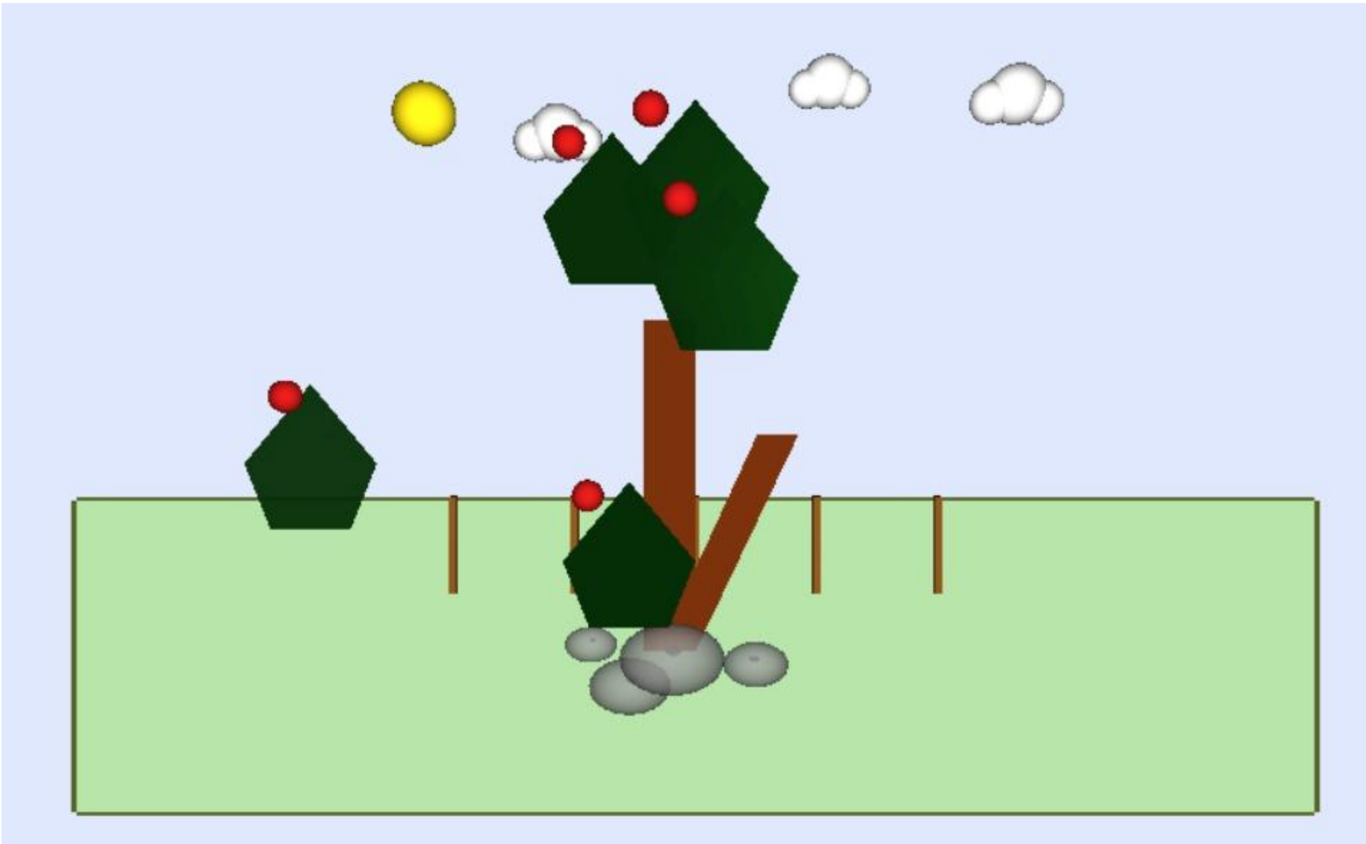
CS360 -Computer Graphics Lab Project
Second semester 1447- 2026



Project Group

Section: 6C9

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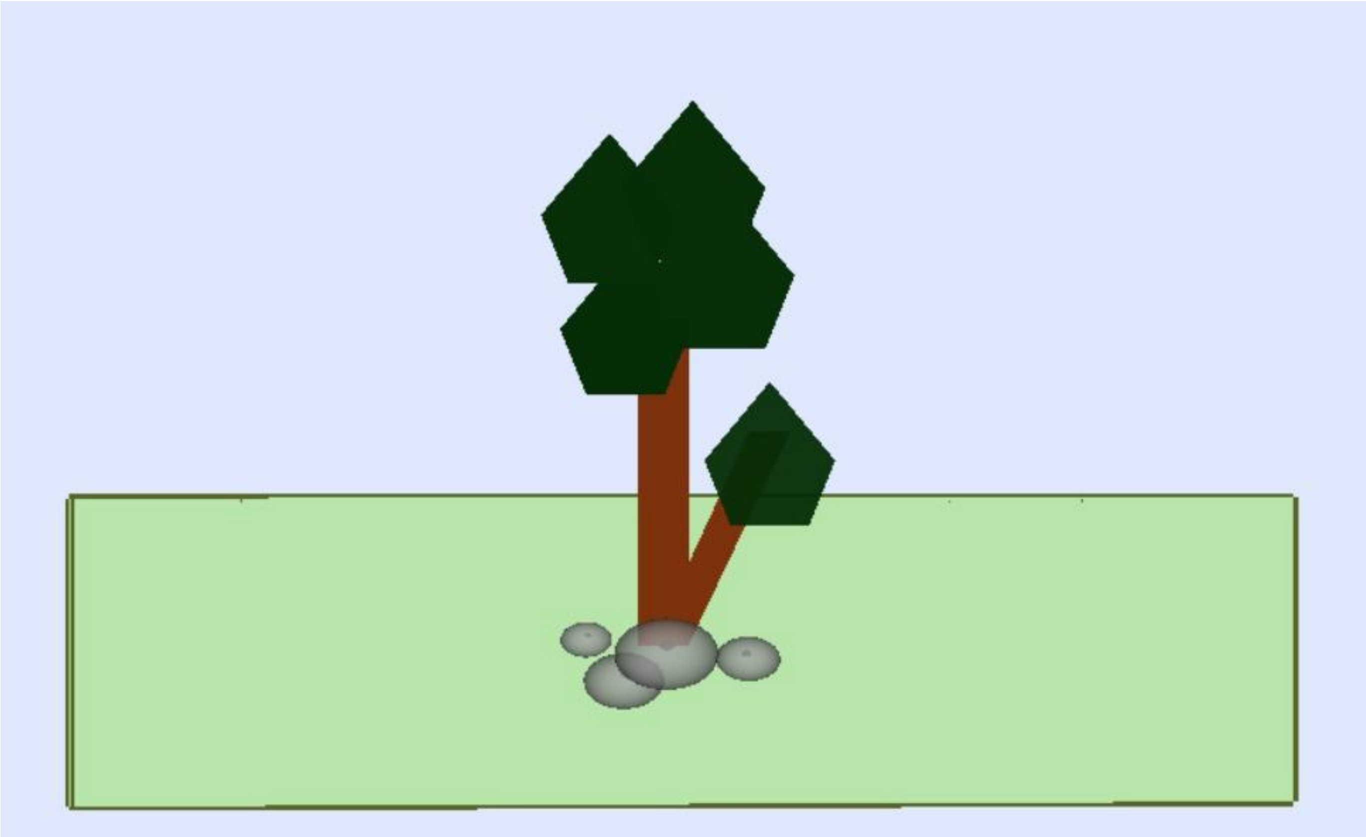
Phase 1: Lighting, depth setup and first sketch



Our design is a vibrant outdoor scene featuring a stylized tree. We built the tree using a brown rectangular trunk and green polygonal leaves, situated on a simple flat grassy base with decorative grey rocks at the bottom to add a natural feel.

Lighting and Depth Setup:

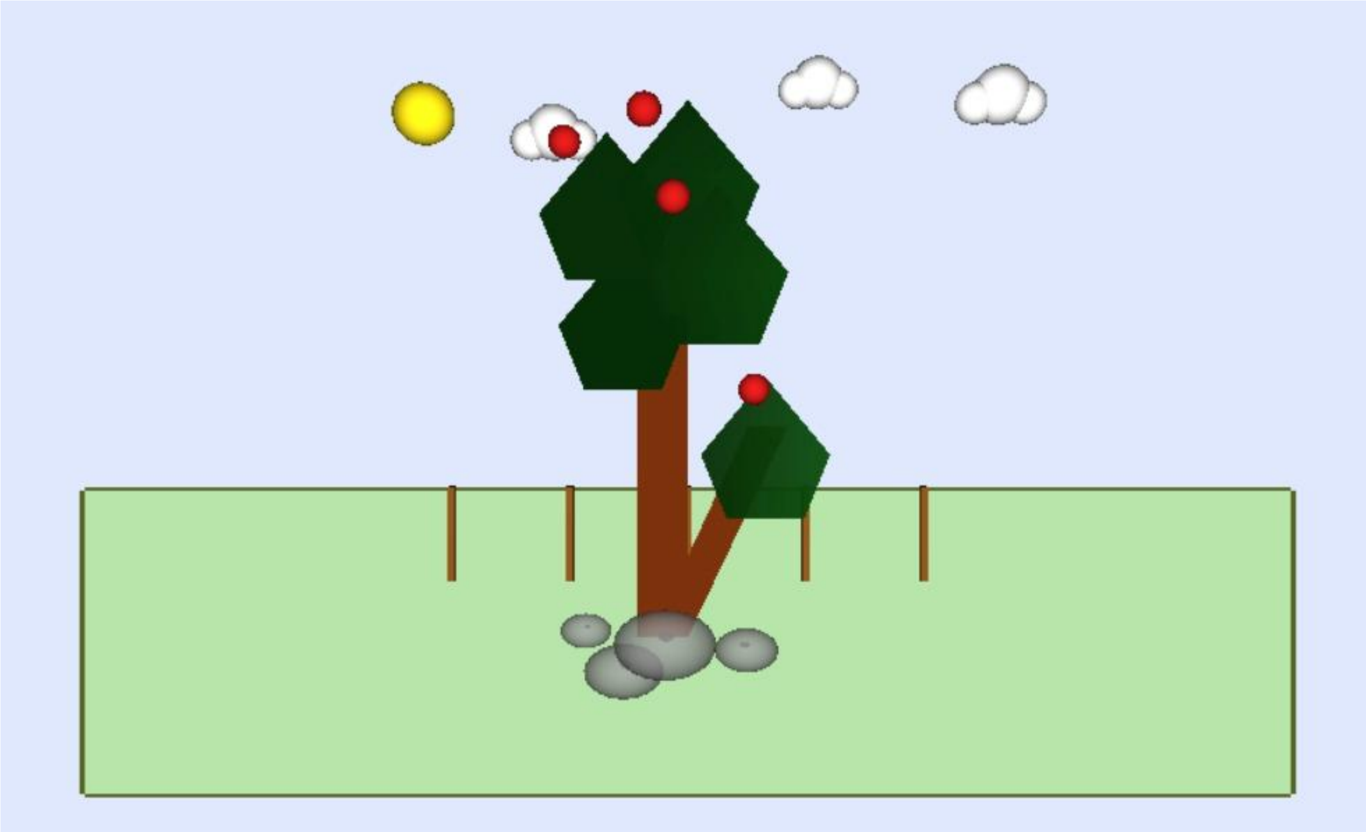
We successfully enabled GL_DEPTH_TEST in the main function to ensure that overlapping objects, like the rocks placed in front of the tree trunk, render correctly based on their Z-axis depth. For lighting, we enabled GL_LIGHTING and GL_LIGHT0, combined with GL_COLOR_MATERIAL to retain the natural colors of our objects. We defined a light position array { 1.0, 1.0, 0.0, 1.0 } to create a realistic 3D shading effect.

Challenges & Solutions:

* Challenge: When we first ran the code, the shapes overlapped incorrectly without depth, and the colors looked flat and washed out due to the light overriding the original colors.

* Solution: We solved the overlapping issue by enabling GL_DEPTH_TEST and adding GL_DEPTH_BUFFER_BIT to our glClear function. We fixed the color issue by correctly enabling GL_COLOR_MATERIAL before applying the light source.

Phase 2: Model Construction



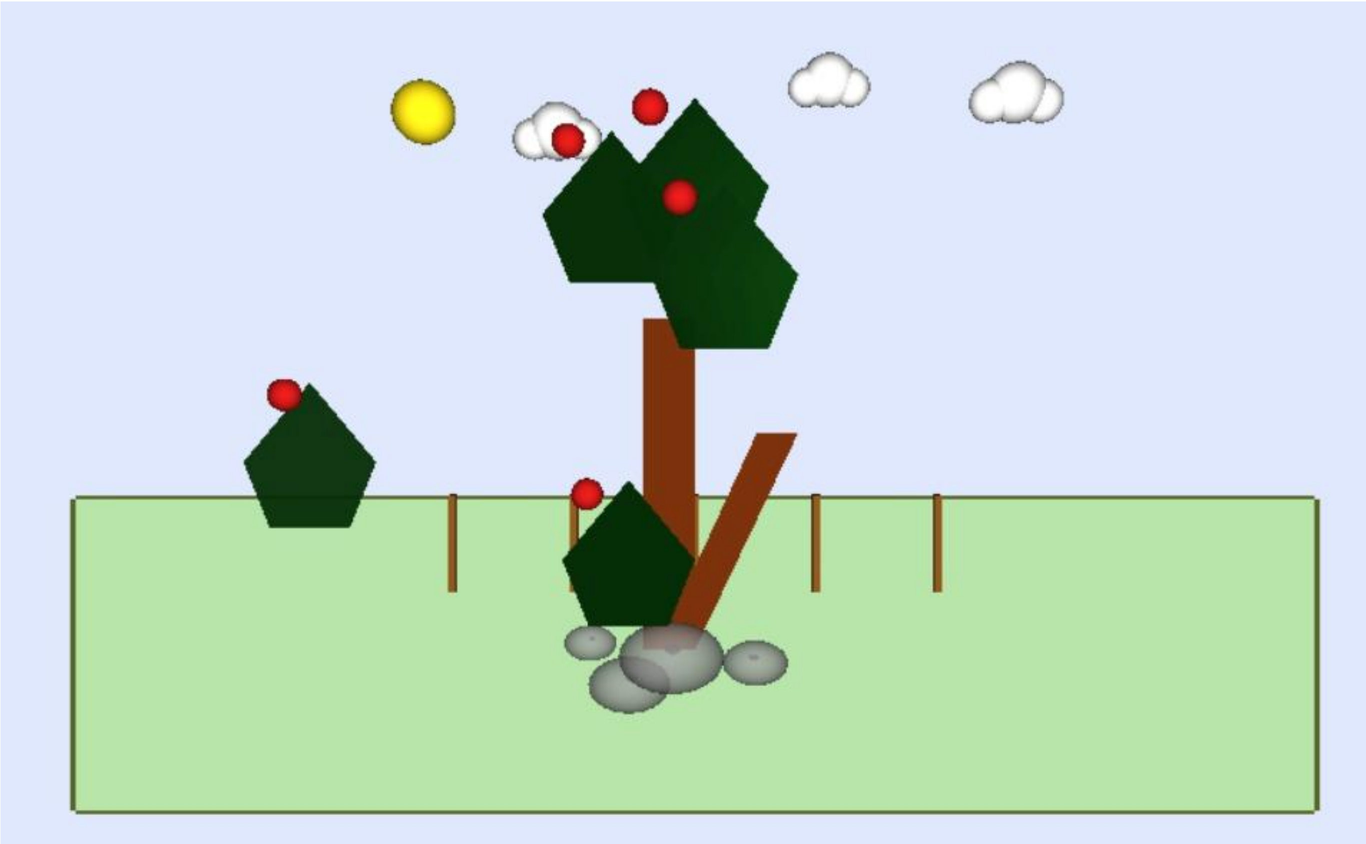
In this phase, we expanded our initial sketch into a fully populated 3D environment, successfully incorporating 24 GLUT objects. We constructed clouds by overlapping multiple glutSolidSphere objects, added a sun using a scaled sphere, and scattered red apples across the tree leaves using spheres. Additionally, we built a background wooden fence using a loop of glutSolidCube primitives. We heavily relied on the transformation stack (glPushMatrix() and glPopMatrix()) to safely position and scale each object independently.

Challenges & Solutions:

* Challenge: Accurately positioning multiple objects (like apples and fence posts) in 3D space was difficult and initially resulted in breaking the coordinates of the main tree.

* Solution: We created modular helper functions (drawApple and drawCloud) to manage complex objects cleanly. For the fence, we implemented a for loop that automatically calculates the correct X-axis offset for each post. We strictly used glPushMatrix() and glPopMatrix() around every object to isolate transformations. What we learned: This experience taught us a lot about spatial awareness in computer graphics and how to use the coordinate system more effectively to organize a complex 3D scene.

Phase 3: Basic Animation & Interaction



* * Animation: We added a dynamic falling animation to one of the apples and a specific leaf. This continuous motion is driven by the dropY variable, updated via glutIdleFunc. We implemented a strict "limit" (if (dropY > -30.0f)) so that the falling objects stop exactly when they hit the ground.

* Interaction: We added user interaction using the keyboard switch controller. Pressing the 'A' or 'B' keys dynamically moves the top leaves left and right by updating the moveX variable. Pressing the 'SPACE' key resets the scene to its default view.

* Final Improvements: We enhanced the background sky by implementing alpha blending using glEnable(GL_BLEND) and glBlendFunc(GL_SRC_ALPHA, GL_ONE_MINUS_SRC_ALPHA), resulting in a softer visual output.

Challenges & Solutions:

* Challenge: The falling objects would drop infinitely off the screen and never stop.

* Solution: We solved the infinite drop by applying a conditional limit statement that halts the subtraction from the animation variable once it reaches the floor coordinates.

Final Reflection

During this project, our team learned how to practically apply the OpenGL rendering pipeline, gaining hands-on experience with the transformation stack, depth testing, and lighting mechanics. The most challenging part was synchronizing the real-time continuous animation with the user-triggered keyboard interactions while keeping the code clean and avoiding rendering bugs. However, the most rewarding part was finally seeing our static 2D-like sketch come alive as an interactive, fully shaded 3D scene. If we had more time, we would improve the project by adding textured materials to the objects instead of solid colors, and perhaps implementing a dynamic day-to-night lighting cycle tied to the animation loop.

Work distribution:

Name	Work
لميس أحمد الفراج	We worked together
شهد فهد الحصيني	We worked together